



Integration

Indefinite Integral

- Integration is the reverse of differentiation.
- If the derivative is known, the original function can be determined.
- e.g: $\frac{dy}{dx} = x^2$ then by inspection we can ask:
 - What function on differentiating will give x^2 ?
 $y = x^3$? $y = \frac{1}{3}x^3$
 - Check the answer by differentiating again.
 - Remember to include integration constant!
 $y = \int x^2 = \frac{1}{3}x^3 + k$
 - Constant is very NB in physics.

Handy Integration Formulas

- $\int bx^n dx = \frac{b}{n+1} x^{n+1} + k$
- $\int \frac{1}{x} dx = \int x^{-1} dx = \ln|x| + k$
- $\int \sin bx dx = -\frac{1}{b} \cos bx + k$
- $\int \cos bx dx = \frac{1}{b} \sin bx + k$
- $\int e^{bx} dx = \frac{1}{b} e^{bx} + k$
- $\int b^{nx} dx = \left(\frac{1}{n} \ln b\right) b^{nx} + k$
- $\int \frac{f'(x)}{f(x)} dx = \ln|f(x)| + k$
- $\int bf(x) dx = b \int f(x) dx$
- $\int [f(x) \pm g(x)] dx = \int f(x) dx \pm \int g(x) dx$

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Examples - 1

- $\int (6x^2 - 8x + 5) dx =$
- $\int (4 \cos 2x) dx =$
- $\int (5 \sin 2x + 3e^{2x}) dx =$
- $\int (\sin^2 x dx) =$

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Definite Integral

- The definite integral of $g(x)$ between the limits $x = a$ and $x = b$ (in this order) is:

$$\int_a^b g(x)dx = [y(x)]_a^b = y(x)\Big|_a^b \\ = y(b) - y(a)$$

- e.g.:

$$\begin{aligned} \int_1^3 (4x-3)dx &= [2x^2 - 3x + k]_1^3 \\ &= [2(3)^2 - 3(3) + k] - [2(1)^2 - 3(1) + k] \\ &= [9 + k] - [-1 + k] = 10 \end{aligned}$$

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Definite Integral

- If the upper and lower limits of the integral are interchanged, the sign of the answer changes:

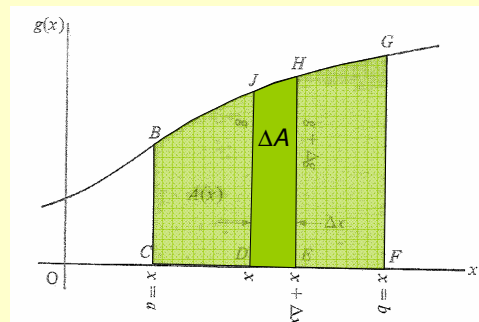
$$\begin{aligned} \int_3^1 (4x-3)dx &= [2x^2 - 3x + k]_3^1 \\ &= [2(1)^2 - 3(1) + k] - [2(3)^2 - 3(3) + k] \\ &= [-1 + k] - [9 + k] = -10 \end{aligned}$$

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Definite Integral and areas

- Consider the figure
- Area under $g(x)$ is BCFG.
- Area
$$\Delta A = \left[\frac{g(x) + g(x + \Delta x)}{2} \right] \Delta x$$
- If $\Delta x \rightarrow 0$
 $g(x) \rightarrow g(x + \Delta x)$
 $dA = g(x) dx$
- Total area under $g(x)$ is given by the sum of all the dA 's:

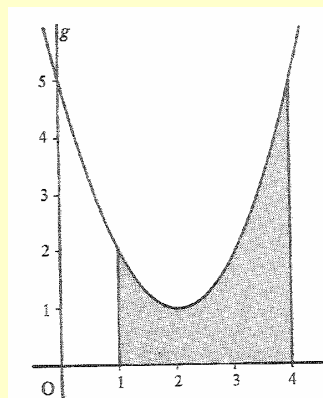
$$A = \int_a^b g(x) dx = [y(x)]_a^b$$



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Example - 2

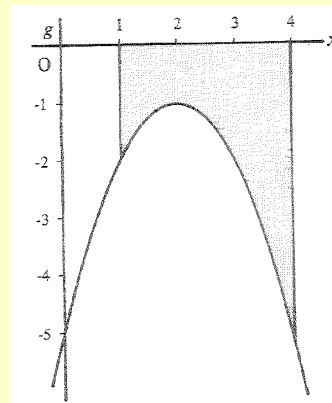
- Calculate the area between the curve $y = t^2 - 4t + 5$ and the t -axis which lies between $t = 1$ and $t = 4$. y represents the gas flow in a pipe in litres per second and t is the time in seconds. See figure.



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Example - 3

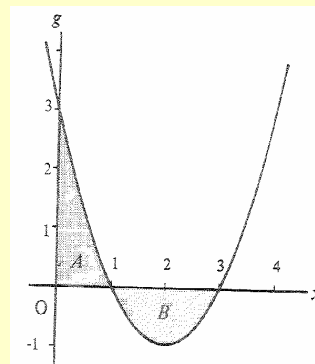
- Calculate the area between the curve $g = -x^2 + 4x - 5$ and the x -axis which lies between $x = 1$ and $x = 4$. See figure.



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Example - 4

- Calculate the area between the graph $y = x^2 - 4x + 3$ and the x -axis between (i) $x = 0$ and $x = 1$, (ii) $x = 1$ and $x = 3$ and (iii) $x = 0$ and $x = 3$. See figure.



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Problems

- If $\frac{dy}{dx}$ is given by the following, determine $y = y(x)$:
 - $3x^2 - 6x + 5$
 - $6e^{3x}$
 - $4x^3 + 3x^2 + 2x + 1 - x^{-1} - x^{-2} - x^{-3}$
 - $4 \sin 2x$
- Determine:
 - $\int (8s^3 - 3 - 4s^{-3}) ds$
 - $\int -18 dx$
 - $\int (v - gt) dt$ v and g are constants
- Determine:
 - $\int_{0^\circ}^{90^\circ} \sin x dx$
 - $\int_1^4 \sqrt{x} dx$
 - $\int_0^{10} x^{-0.6} dx$